RAG Technology Lab · Architectural Overview

Retrieval-Augmented Generation (RAG) is a pattern that augments Large Language Models (LLMs) with verified information retrieved dynamically from an external knowledge base.

Key Pipeline Steps:
1. Document Ingestion: Parse PDFs, OCR scans, and extract text/tables.
2. Chunking & Embedding: Split text into overlapping chunks and generate vector embeddings.
3. Vector Indexing: Store embeddings in PostgreSQL pgvector or dedicated vector stores.
4. Semantic Search: Perform cosine similarity matching for inbound user queries.
5. Generation & Citation: Provide grounded answers accompanied by exact page citations.

Component	Selected Technology	Primary Purpose
Vector Database	PostgreSQL + pgvector	Vector Index & Similarity Search
Embeddings	BGE-M3 / Multilingual-E5	Multilingual Representation
Local LLM	Ollama / Qwen2.5-7B	Grounded Multilingual QA

Multilingual Processing & Khmer Support

Section 2: Multilingual Document Ingestion

Processing non-Latin scripts such as Khmer requires Unicode normalization (NFC) and specialized multilingual embedding models like BGE-M3 to capture semantic similarity accurately.

Sample Reference Text (Khmer):
???????? RAG ?????????????????????????????? AI ????????????????????????
????????????????????????????????

Metadata:
- Target Script: Khmer (U+1780 to U+17FF)
- OCR Engine: Tesseract with 'khm' and 'eng' traineddata
- Vision Model: Qwen2.5-VL for document chart and diagram reasoning.